CREATING A PLAYGROUND TACTILE MAP

LSU Biological Engineering Capstone 2025-2026

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Playground Tactile Map Creation Protocol

1. Purpose

This document provides a step-by-step protocol for generating a tactile playground map using photogrammetry, geometric refinement, feature extraction, tactile encoding, and 3D fabrication. This protocol was developed as an Engineering Capstone Project at Louisiana State University in the 2025-2026 academic year. The purpose regards children with visual impairments, focusing on accessibility to play equipment. Providing a tactile map of a playground for a child increases confidence and independence during play, allowing these children to navigate their environment in a safe way.

For maximum repeatability, the materials used in the 2025-2026 capstone project are listed, along with other options that are budget-friendly or easier to acquire. Steps are listed in order, with a “best practice” section for each main step to help with troubleshooting, and a following numbered protocol to guide you through the process.

Examples of work after each step are included, which were captured from the 2025-2026 capstone project. They are to serve as a guide, as well as an attached video demonstration that shows the greater steps of the process.

2. Overview of Workflow

The workflow consists of three major domains:

Acquisition – Collecting high-quality images and video frames.
Processing – Refining geometry, extracting features, and encoding tactile elements.
Deployment – Fabricating, validating, and installing the tactile map.

PROJECT MATERIALS

Acquisition:

- DJI Mini Pro 4 drone (48 MP, 4K/60 FPS), supplemented by smartphone photos and videos taken on an iPhone 17 Pro (48 MP, 4K/60 FPS)
  - Most handheld smart devices would work for this step, especially if it can capture high-quality media.
- 3DF Zephyr photogrammetry software
  - Other software can be used, but it must output files in a supported 3D model file for ease of 3D printing

Processing:

- Blender
  - Blender is a free, open-source program used to further process the models.
- Other processing tools used were created by the 2025-2026 capstone group and are managed by the Community Playground Project at Louisiana State University. These resources can be accessed through their website.

Deployment:

- A kiosk for the tactile map that shields it from the elements, keeping in mind that the map will be used primarily by children.
- 3D printer
- PLA or chosen material, like high-density polyethylene

SECTION A — ACQUISITION

A1. Image & Video Collection Procedure

In this section, media will be collected to begin creating the 3-D model of the chosen playground.

Best Practice

- Consistent lighting will yield a higher-quality result
- Avoid motion blur in photos
- There should be overlap maintained between photos, around 70-80% if possible.
- Videos should be captured if possible and are especially helpful for continuity of the future 3D model.
    - 4K videos are preferred, with a 60FPS capture speed. Frames can later be extracted for modeling.

Procedure

1. Use chosen photogrammetry method to capture high-resolution media of the entire playground, from different angles and heights.
    a. The device should travel all the way around the circumference of the playground, capturing as many different vantage points of the playground as possible.
        i. When traveling around the outside of the playground, try to maintain even distance from the playground elements all the way around.
    b. The device should also capture media from a simulated walking path on the platforms of the playground, to properly convey stairs, ramps, and platforms from aerial angles.

A2. 3DF Zephyr: Dataset Upload

This section takes playground media and compiles it into a preliminary 3D model.

Best Practice

- Use automatic calibration settings
- Always upload photos first, then videos to reduce processing time
- Total dataset size should be at least 500-1300 images/frames

- Dataset can include less if the images are high-quality and maintain high overlap

Procedure

1. Open 3DF Zephyr and start New Workflow
2. Select Automatic Calibration
3. Load new images first, then video frames.
4. Extract frames from videos if needed
    a. 1 or 2 fps works best for frame extraction.
5. Ensure Data size is at least 500-1300 images/frames

A3. 3DF Zephyr: Sparse Point Cloud (SPC) Generation

This section generates the Sparse Point Cloud. The resulting 3D combination of uploaded data represents the overlap of camera position and analysis of any missing detail in the capture of the playground.

Following data import and calibration, 3DF Zephyr will generate the first and arguably the most important component of a reconstruction model, the Sparse Point Cloud. This will appear as a low-density 3D representation of the target structure that lays out a foundation for later processing steps.

Best Practice

- Use the blue boxes shown in 3DF Zephyr to analyze the amount of media coverage you have from all angles of the playground; it is easier to tell coverage before the more complicated Dense Point Cloud (DPC) Model is created.

Procedure

1. 3DF Zephyr uses the data uploaded to generate a sparse point cloud. This is a low-density 3D representation of the playground that will be further processed in other steps. This is the foundation of a good tactile map, and having a comprehensive data package to generate the SPC is integral.
2. Inspect camera positions for overlap and as much detail coverage as possible. If there are large gaps in media, upload additional images or frames to supplement the missing sections.
    a. Each camera angle is represented by a blue box in the point cloud space, and can be used to inspect the camera positions, overlap quality, and regions of the playground with missing detail capture.

    b. Additional photos or video frames can be uploaded to fill in any major gaps, but later processing steps will smooth textures and fill in minor holes.
3. Use the **Bounding Box** tool to remove background “noise” from the model, in the form of large extraneous elements (people, buildings, or other structures, etc.). This reduces processing time in future steps and highlights the structures that the software should focus on.
4. Steps 1-3 can be repeated as more media information is added between iterations of model creation, until a consistently dense media coverage is achieved.

A4. 3DF Zephyr: Dense Point Cloud & Mesh Generation

The Dense Point Cloud (DPC) and mesh generation procedures begin to construct the overall model that will be exported for the later post-processing steps. Structural geometry and rough texture details begin to appear on the model, and any inconsistencies are made obvious. The DPC portion itself is usually edited very little, but it is the foundation for creating the mesh. The mesh is the most significant component in the overall outlook of this project

Best Practice

- The DPC is not edited very much, but the resulting mesh will be.
- Each mesh filter applied generates a new iteration of the model to allow visualization of what each filter does. This should be used to isolate filter steps and make sure the desired result is achieved from each one, with correction applied as needed.
- Generating a textured mesh is not necessary for the scope of this project, as it does not impact the geometric accuracy of the mesh, and will significantly increase processing time.

Procedure

1. Generate the Dense Point Cloud (DPC), which is the precursor to the final mesh used to make the physical 3D model of the playground.
2. Generate the mesh
3. Apply mesh filters:
    a. Photo-consistency optimization
    b. Decimation
        i. Greatly reduces vertex count, yielding a model that is more accurate to the physical reality.

    c. Watertight hole-filling
        i. Helps fill in minor gaps in image data to create a smoother and more accurate model surface.
4. Export/save model as a .stl, .obj, or other supported 3D file type to best suit the post-processing printing procedures and to maintain reliable communication between the softwares used in this project.

SECTION B — PROCESSING

After the initial 3D rendering of the playground is created, it must be edited to yield a final model that can then be printed as a usable, accurate tactile map.

B1. Hole Filling (Blender)

Blender will be used to patch holes in the mesh that are not present in the physical playground.

Best Practice

- These steps are used to fill holes that look similar to the ones below. These holes represent missing data in a place where there should be a consistent texture and remediate such issues.

Procedure

1. Import mesh into Blender
2. Identify a hole:
    a. Navigate to the modeling tab of Blender, which shows the model with vertices and faces, and zoom in/identify the first hole to be filled.
3. Select the area surrounding the hole
    a. Will show selected vertices and faces as highlighted
4. Select the fill tool to create new faces across the hole, resulting in a consistent surface.

B2. Feature Extraction - Detecting and confirming platforms

Procedure

Detect Platforms

1. Upload the scanned playground model into the interface.
2. Select **Run Detect Platforms** to initiate automated platform detection.
3. Allow the system to process the scan; a loading interface will display during computation.
4. Review the detected platforms displayed in the 3D viewer. Note that boundaries may appear uneven or contain gaps due to direct extraction from the scan.
5. Click on any detected platform to open the **Platform Properties** panel in the lower-right corner.
6. For each platform, adjust properties as needed:
    - **Change Platform Type** (ground, deck, ramp).
    - **Split Platform** by selecting points across the surface to define a vertical splitting plane.
    - **Delete Platform** if it was incorrectly identified.
7. Continue refining all platforms until the set accurately reflects the intended geometry.
8. Select **Run Smooth Boundaries** to generate smoothed platform outlines.
9. Wait for processing to complete; the smoothed boundary model will appear in a 3D viewer.
10. Once satisfied with the smoothed results, click **Continue** to proceed.

Confirm Platforms

11. Verify that all simplified platforms and slides from the previous step have loaded into the 3D viewer.
12. Select any platform to view its vertices, sides, and faces.
13. Modify geometry as needed:
    - **Move Vertices** individually or as a selected group.
    - **Delete Vertices** using the properties panel; adjacent vertices will automatically form a new side.
    - **Split Sides** to insert additional vertices.

- **Move, Rotate, or Delete Entire Platforms** if repositioning is required.

14. Adjust slide geometry similarly: move, rotate, or delete slides as needed.
15. Enable the **Original Scan Overlay** option to align platforms precisely with the underlying scan.
16. Inspect all platform boundaries and ensure adjacent elements connect cleanly without gaps.
17. When the platform geometry matches the desired final layout, select **Apply and Continue Build**

B3. Tactile Encoding

18. Upon entering the **Add Textures** tab, review the fully assembled 3D model containing extruded platforms and slides.
19. Observe that all detected platforms are highlighted in distinct colors for easy selection.
20. Click on any highlighted platform to open the **Texture Properties** panel in the lower-right corner.
21. Choose a surface type from the dropdown list to assign a texture to the selected platform.
22. To add missing platforms, select any unhighlighted face and assign it as a platform.
23. To remove incorrectly highlighted platforms, select the platform and remove its designation.
24. Continue assigning textures until all platforms have the desired surface types.
25. Select **Generate Textured Model** to produce the final textured 3D model.
26. Wait for the generation process to complete; the textured model will appear in the 3D viewer.
27. Use the download options above the viewer to export the model in the desired file format.

SECTION C — DEPLOYMENT

C1. Printing and Installation

Best Practice:

- If not using a 3D printer that can print the entire map in one piece, be mindful of the way that you attach the pieces together. The research group chose to implement “hooks” into the sides of the cut print to slide together. Adhesive can also be used but make sure it is compatible with the type of filament used.

Protocol:

1. Gather 3D printing files in a format that will work with the 3D printing software being used.
2. If print needs to be split, use the slicer for the 3D printer chosen, and separate model into pieces, being mindful of how they will connect.
3. After printing, install map in kiosk, being mindful of weather, UV exposure, and ease of use for children in kiosk construction.

End of Manual